
CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Roberts, Jesse

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-6210-0678>

*POSITION TITLE: Assistant Professor

*ORGANIZATION AND LOCATION: Tennessee Technological University, Cookeville, Tennessee, United States

Proposals and Active Projects

*Proposal/Active Project Title: Wilson Project

*Status of Support: Current

Proposal/Award Number:

*Source of Support: Wilson Sports

*Primary Place of Performance: Tennessee Tech University

*Proposal/Active Project Start Date: (MM/YYYY): 01/2026

*Proposal/Active Project End Date: (MM/YYYY): 12/2026

*Total Anticipated Proposal/Project Amount: \$39,832

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2026	0.1

*Overall Objectives:

This project is researching methods for automatic image vectorization to support manufacturing.

*Statement of Potential Overlap:

No known potential overlap

***Proposal/Active Project Title:** Collaborative Research: Conference: NAIRR Pilot Expansion: FA1: Mid-South Conferences on CyberInfrastructure Advances to Enable Interdisciplinary AI Research in Health Disparities

***Status of Support:** Current

Proposal/Award Number: 2526362

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Tennessee Technological University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 12/2026

***Total Anticipated Proposal/Project Amount:** \$125,133

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.29
2026	0.13

***Overall Objectives:**

The goal of the proposed conference series is to provide training, guidance, and support to the formation and furtherance of interdisciplinary research at the nexus of Advanced CyberInfrastructure, health disparities, and the domain sciences underlying the new and innovative solutions, including AI (deep learning, LLMs, etc.), which is a timely and vital area that needs to be advanced for the good of a large portion of the population in the Southeastern US and beyond.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** ERI: Data-Driven Expansion of Electrified Transportation Network

***Status of Support:** Current

Proposal/Award Number: 2347337

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Tennessee Technological University

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2024

***Proposal/Active Project End Date: (MM/YYYY):** 05/2027

***Total Anticipated Proposal/Project Amount:** \$195,769

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1

***Overall Objectives:**

In this proposal, we plan to investigate the long-term ETN expansion strategy considering scalable EV service forecasting model and long-term LULC change impact.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** CICI: IPAAI: Provenance-Aware AI Workflows: Building Trust and FAIR-ness in AI Datasets and Models

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation (NSF)

***Primary Place of Performance:** Tennessee Technological University

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2026

***Proposal/Active Project End Date: (MM/YYYY):** 09/2029

***Total Anticipated Proposal/Project Amount:** \$900,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

This project aims to develop an opt-in, blockchain- and smart-contract-based cyberinfrastructure that enhances the integrity and provenance of AI datasets and models. The system will support provenance-aware registration, verification, and large-file transfer within Python-based AI workflows, while integrating with major repositories such as Hugging Face, Kaggle, and Mendeley Data. By

enabling end-to-end provenance tracking, model-dataset linkage, community feedback, and post-hoc impact analysis, the project aims to enhance trust in AI-enabled scientific results and promote provenance enhanced workflows as a community best practice.

***Statement of Potential Overlap:**

This project has no apparent overlap with other projects.

***Proposal/Active Project Title:** Cherokee beneath the surface: Does morphological parsing permit better neural machine translation and analysis of etymology?

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Schmidt Sciences

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2026

***Proposal/Active Project End Date: (MM/YYYY):** 07/2028

***Total Anticipated Proposal/Project Amount:** \$297,752

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2
2027	2

***Overall Objectives:**

The goal of this project is to build on our state of the art Cherokee translation model and train an LLM to use a novel intermediate Cherokee form that is expected to dramatically boost performance and help answer linguistic questions regarding Cherokee word constructions.

***Statement of Potential Overlap:**

No known overlap

***Proposal/Active Project Title:** Collaborative Research: GUARDIAN: Generative AI for Unified Augmented Risk Detection, Intervention, and Analysis in Networked Power Systems

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation (NSF)

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2026

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** \$399,999

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.71
2028	0.71
2029	0.71

***Overall Objectives:**

The project proposes GUARDAIN, which relies on multi-modal data and context-aware reasoning to predict the future system state, the impact of potential cyber-physical vulnerabilities, overall risk to continued operation, and generates an actionable mitigation plan informed by operator preferences and physical limitations through LLM reinforcement learning.

***Statement of Potential Overlap:**

There is no apparent potential overlap in the projects.

***Proposal/Active Project Title:** AI-Driven Observable Synthesis for the INQ
Electronic Structure Engine via Large Language
Models: Accelerated Materials Discovery for the
Genesis Mission

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Department of Energy

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	0.5

***Overall Objectives:**

This project seeks to use AI to convert natural language statements of relationships into precise hypotheses using LLMs. Those relationships will then be converted to precise and performant HPC code by a separate LLM process. I will be leading the LLM application development.

***Statement of Potential Overlap:**

There is no anticipated overlap

***Proposal/Active Project Title:** PRISM: Process Intelligence for Rapid, Integrated SOFC Manufacturing

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Department of Energy

***Primary Place of Performance:** Tennessee Tech University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5

***Overall Objectives:**

This project seeks to apply AI to the manufacturing of solid oxide fuel cells to optimize the process variables by developing a structural digital twin by learning the mapping between process variables and molecular structure.

***Statement of Potential Overlap:**

There is no known potential overlap with other work.

***Proposal/Active Project Title:** Collaborative Research: NeTS: Compact Representations for Exascale Advanced Measurements (CREAM)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation (NSF)

***Primary Place of Performance:** Tennessee Technological University

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2026

***Proposal/Active Project End Date: (MM/YYYY):** 05/2029

***Total Anticipated Proposal/Project Amount:** \$930,552

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1
2028	1

***Overall Objectives:**

This project proposes a new representation of network traffic that makes large-scale, multi-site analysis practical. This project aims to do this generating tokens at capture time and transforming them into embeddings and models that can be retained and reused at scale. Instead of collecting raw traces for offline processing, traffic is tokenized as it is captured, preserving ordering and behavioral structure in a compact form. Token sequences are then mapped into learned embeddings and used to train local models at distributed vantage points. These representations, tokens, embeddings, and models, can be merged across sites to support system-wide analysis and revisited later to answer new semantic questions without building one-off analysis pipelines.

***Statement of Potential Overlap:**

There is no known overlap between this and other projects.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Roberts, Jesse in SciENcv on 2026-04-27 21:26:23